Hao Chen

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Personal Data

Date of Birth: 3 Mar, 1997
Place of Birth: Jiangsu Province, China
Gender: Male
Nationality: China
Language: Chinese and English

Research interests

Extremal Combinatorics, Probabilistic Combinatorics and Graph Limits

Academic appointments

2026.7 - present **Lecturer**
School of Mathematical Sciences, Soochow University.

Education

2019.9 - 2026.6 **Ph. D. in Applied Mathematics**
School of Mathematical Sciences, University of Science and Technology of China,
Supervisor: Prof. Jie Ma

2015.9 - 2019.6 **B. Sc. in Mathematics and Applied Mathematics**
School of Mathematical Sciences, Soochow University
Supervisor: Prof. Lijun Ji

2024.11 - 2025.11 **Visiting Ph. D. Student in Mathematics**
Department of Mathematics and Statistics, University of Victoria,
Supervisor: Prof. Jonathan A. Noel

Research Papers

- A property on monochromatic copies of graphs containing a triangle (with Jie Ma). *SIAM J. Discrete Math.* **38(1)** 2024, 316–326.
- Undecidability of polynomial inequalities in tournaments (with Yupeng Lin, Jie Ma and Fan Wei). *Int. Math. Res. Not.* (IMRN) **2025(11)** 2025, rnaf131.
- On alternating 6-cycles in edge-coloured graphs (with Jonathan A. Noel). *Electron. J. Combin.* Volume 33, Issue 3 2026, #P3.27.
- Kohayakawa-Nagle-Rödl-Schacht conjecture for subdivisions (with Yupeng Lin and Jie Ma). *SIAM J. Discrete Math.*, to appear.

- Maximizing alternating paths via entropy (with Felix Christian Clemen and Jonathan A. Noel). *arXiv:2505.03903.*
- On tournament anti-Sidorenko orientations of trees (with Felix Christian Clemen and Jonathan A. Noel). *arXiv:2605.14138.*
- Oriented paths with few direction flips are tournament anti-Sidorenko (with Yupeng Lin). *arXiv:2609.14655.*
- Elementary proofs of commonality of the house graph and K_4 with one edge subdivided (solo paper). *arXiv:2610.20149.*

Teaching Experience

- **Extremal Combinatorics**, University of Science and Technology of China, Teaching Assistant, Spring 2023.
- **Extremal Problems in Graphs and Hypergraphs**, Tianyuan Mathematical Center in Southwest China, Teaching Assistant, July 2024.
- **Topics in Graph Theory**, University of Science and Technology of China, Teaching Assistant, Spring 2026.
- **Calculus I**, Soochow University, Fall 2026.

Talks

- **Some New Results on Common Graphs and Strongly Common Graphs**, USTC Workshop on Combinatorics 2023. Hefei, China. August 2023.
- **On Problems of Inequalities in Homomorphism Densities – Sidorenko's Conjecture and KNRS Conjecture**, The 11th Chinese Combinatorics and Graph Theory Conference. Qingdao, China. September 2024.
- **On Problems of Inequalities in Homomorphism Densities – Sidorenko's Conjecture and KNRS Conjecture**, Discrete Math Seminar at UVic. Victoria, Canada. November 2024.
- **Undecidability of Polynomial Inequalities in Tournaments**, Canadian Discrete and Algorithmic Mathematics Conference 2025 (CanaDAM 2025). Ottawa, Canada. May 2025.
- **Semi-Inducibility of Alternating Paths and Cycles**, SCMS Combinatorics Seminar. Shanghai, China. December 2025.
- **Semi-Inducibility of Alternating Paths and Cycles**, USTC Winter Workshop on Combinatorics and Graph Theory. Hefei, China. January 2026.
- **Counting Graph Homomorphisms via Entropy**, CSIAM 2026 Academic Conference of the Graph Theory, Combinatorics and Applications. Shijiazhuang, China. May 2026.
- **On Tournament Anti-Sidorenko Orientations of Trees**, The 12th Chinese Combinatorics and Graph Theory Conference. Zibo, China. July 2026.